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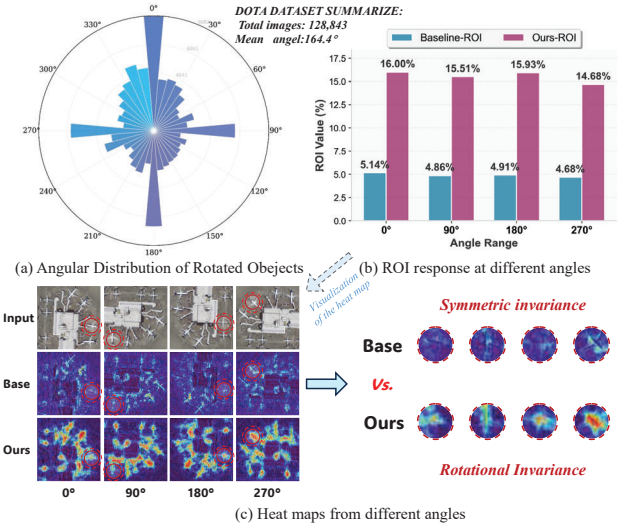
Rotation Invariant and Symmetry Aware Pixel Difference Network for Remote Sensing Object Detection

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Project page: https://github.com/yuhua666/RIS-PiDiNet

Abstract

Recent advancements in remote sensing object detection have predominantly focused on oriented bounding box design and small object feature enhancement, while often overlooking the intrinsic geometric properties of remote sensing images, such as rotation invariance and structural symmetry. Many aerial objects appear in multiple orientations and exhibit clear symmetrical patterns, which, if not explicitly modeled, can lead to detection failures and inaccurate localization under geometric variation or partial occlusion. To address this, we propose the Rotation Invariant and Symmetry Aware Pixel Difference Network (RIS-PiDiNet), which introduces a novel convolutional operator called Rotation Invariant and Symmetry Aware Pixel Difference Convolution (RIS-PDC). This operator replaces traditional convolution with a mathematically grounded formulation that encodes rotation group priors and symmetrical constraints. RIS-PDC utilizes pixel differences and symmetry-guided aggregation in the polar harmonic space, enabling the network to infer partially visible structures and deduce occluded symmetrical parts. Besides improving detection accuracy, RIS-PDC enhances model interpretability by embedding geometric principles into the network design. Feature visualizations demonstrate rotation-consistent activations and symmetry-complete responses, revealing how the network captures underlying object structure even under partial visibility or orientation changes. This yields geometrically interpretable detection decisions. To our knowledge, RIS-PiDiNet is the first remote sensing object detection framework that jointly incorporates rotation invariance and symmetry modeling within a unified architecture. Extensive evaluations on standard benchmarks validate its effectiveness, achieving state-of-the-art performance on DOTA-v1.0 (78.53% mAP single-scale, 81.81% multi-scale), HRSC2016 (98.60% mAP), and DIOR-R (67.28% mAP), all with acceptable computational overhead and no increase in parameter count.



(a) Angular Distribution of Rotated Obejects (b) ROI response at different angles

(c) Heat maps from different angles

Figure 1. Geometric analysis of rotation invariance and symmetry in remote sensing. (a) Angular distribution of rotated objects in DOTA-v1.0, showing non-uniform density across orientations. (b) Quantitative ROI responses at different rotation angles: our RIS-PiDiNet maintains stable activation intensity across all directions, while the baseline exhibits high variance indicative of orientation bias. (c) Qualitative heat map comparisons confirm that our model preserves rotation-invariant and symmetry-consistent activations.

1. Introduction

In remote sensing applications such as land use classification, urban planning, and environmental monitoring [20, 49, 70], object detection plays a strategic role. The human visual system possesses a remarkable ability to recognize objects under arbitrary rotations and partial occlu-

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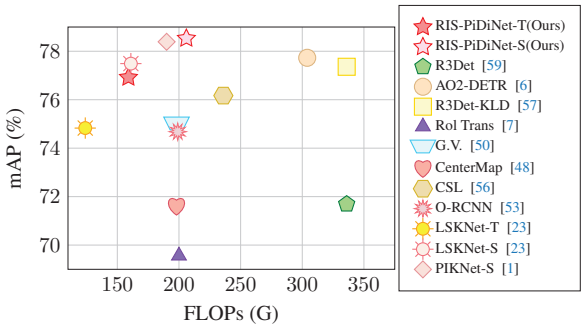


Figure 2. A comprehensive evaluation of the detection performance of different remote sensing object detectors on the single scale of DOTA dataset.

sions by leveraging symmetry and local structural consistency. However, such geometric sensitivity remains difficult to replicate in current deep models. Specifically, the diverse orientations of objects in aerial imagery (Figure 1(a)) and their non-uniform angular distribution pose a major challenge for achieving stable feature responses under rotation. As shown in Figure 1(b), existing models exhibit strong bias toward specific orientations [70], leading to inconsistent ROI activations across angles. This angular sensitivity reflects a deeper misalignment between the geometric priors of rotated objects and the structural representations learned by convolutional networks. In contrast, our proposed RIS-PiDiNet demonstrates both rotation invariance and symmetric consistency (Figure 1(c)), maintaining uniform responses and balanced feature activation patterns across multiple orientations, thereby bridging the gap between visual geometry and neural representation.

Two dominant approaches have emerged: rotation-aware detection, using angle-regression with orientation-sensitive loss [24, 31, 39, 58], and rotation-invariant modeling, using equivariant architectures or augmentations to reduce feature disparity [18, 19, 34, 35, 43, 46, 73]. However, both remain fragile under extreme viewpoints or symmetric object layouts, where angle sensitivity destabilizes boxes and invariant features blur mirror symmetries. These limitations raise two fundamental questions: (i) Does rotation invariance truly preserve the continuous SO(2) geometry of orientation space? (ii) How can symmetry, an intrinsic structural prior, be explicitly encoded to stabilize feature learning?

To bridge this gap, we propose RIS-PiDiNet, a rotation-invariant and symmetry-aware pixel difference network that integrates classical geometric theory into deep representation learning. First, RIS-PiDiNet embeds continuous SO(2) rotation [10, 17] directly into convolution, ensuring directional coherence in feature extraction. Second, it introduces a Polar Harmonic Transform (PHT) [63] to capture structured polar frequency features of symmetric objects. Third, it enhances Convolution with Local Binary Patterns (LBP) [37, 44, 67] to better express local consistency (Figure 1(b)-(d)). Together, these design elements allow RIS-PiDiNet to model geometric structure more faithfully, yielding significant gains in detecting rotated and symmetric objects without sacrificing computational efficiency (Figure 2).

In summary, the contributions of our work are threefold:

- We propose RIS-PiDiNet, the first remote sensing detector to unify rotation invariance and internal structural symmetry within a single geometry-driven representation framework.
- We introduce RIS-PDC, a mathematically grounded convolution that embeds SO(2) rotation priors and harmonic symmetry constraints, enabling stable and symmetry-aware feature extraction.
- The proposed modules plug seamlessly into mainstream detectors and deliver consistent gains across DOTA-v1.0 (+1.04% mAP), HRSC2016 (+1.38%), and DIOR-R (+0.14%), with especially notable improvements on challenging categories.

2. Related Work

Remote Sensing Object Detection. ***Two-stage methods.*** RoI Transformer [7] rotates anchors for better alignment; [55] and [58] refine vertices or fuse multi-layer features for small objects. [68] combines global and local features, Oriented RCNN [53] stabilizes training losses, and LSKNet [23] enlarges context via large kernels. ***Single-stage methods.*** RetinaNet-based [26] works include [38] with modulated loss and OHDet [56] with circular smooth labels. S2ANet [12] aligns features directionally; R3Det [59] refines via progressive regression. ***Transformer-based methods.*** AO2-DETR [6] improves positional priors; ARS-DETR [66] uses rotation-deformable attention; EMO2-DETR [15] mitigates mislabeling; QETR [29] preserves local features. Although these methods improve angle and feature handling, they fall short in capturing multi-directional response consistency. Our approach leverages group theory and classical domain insights for robust orientation-aware detection.

Rotation Invariance and Symmetry. Group-equivariant convolutions [5] and Oriented Response Networks [71] model rotations but often at high cost. ReDet [13], FRNFormer [46], and R2FD2 [72] embed invariant/equivariant mechanisms yet face efficiency trade-offs. Symmetry-aware works (R2Det [51], LodeSTAR [30], Reflect3D [22]) indicate symmetry modeling's potential. Unlike architecture-level approaches (e.g., ReDet) that rely on heavy feature realignment, our RIS-PDC is a plug-and-play operator that embeds SO(2) priors directly into kernel weights. Our method fuses SO(2) rotation, polar harmonic transform and LBP operators within differentiable convolutions to achieve efficient, rotation invariant and symmetry aware modeling without altering

the detection framework.

3. PROPOSED RIS-PiDiNet

Remote sensing objects exhibit arbitrary rotation (overhead viewpoints) and intrinsic structural symmetry (e.g., bilateral aircraft, radial roundabouts). Standard convolutions lack mechanisms for either property. The following subsections develop a unified operator encoding symmetry awareness (S-PDC) and rotation invariance (RIS-PDC). We propose RIS-PiDiNet, a geometry-consistent detector designed for remote sensing object detection under arbitrary rotations and structural symmetries. In contrast to prior methods that rely on large receptive fields for contextual aggregation [23], RIS-PiDiNet explicitly integrates geometric priors into feature learning through two complementary modules: rotation modeling based on Lie Group theory [10] and symmetry modeling guided by the Polar Harmonic Transform [63]. These modules are jointly implemented within a unified convolutional operator, RIS-PDC, which enhances local texture sensitivity and preserves geometric consistency across orientations. The overall architecture (Figure 3) follows a stacked RIS-block design inspired by LSKNet [23], augmented with lightweight local descriptors such as LBP [37] to further strengthen fine-grained structural representation.

3.1. Symmetry Aware Pixel Difference Convolution (S-PDC)

Before introducing our symmetry-aware extension, we briefly revisit the Pixel Difference Convolution (PDC) [44, 67]. Given an $N \times N$ input patch $\boldsymbol{x}$, PDC replaces the raw pixel aggregation in vanilla convolution with local differences to a reference value q:

$$y = \sum_{i=1,\, i\neq c}^{N^2} w_i \,(x_i - q), \tag{1}$$

where x_i denotes the intensity at the i-th location in the patch, c is the index of the central pixel, w_i is the kernel weight, and q is a reference value (e.g., the central pixel or a constant). Following [44, 65], we adopt a fixed $q = 0.2$ in practice for numerical stability.

To explicitly encode structural symmetry, we introduce Symmetry-aware Pixel Difference Convolution (S-PDC), which modulates PDC with symmetry-sensitive harmonic kernels derived from the Polar Harmonic Transform (PHT) [63]. For a single harmonic order (n, l), the S-PDC response is defined as

$$y^{(n,l)} = \sum_{i=1,\, i\neq c}^{N^2} w_i \, H_i^{(n,l)} \,(x_i - q), \tag{2}$$

where $H_i^{(n,l)}$ is the harmonic kernel evaluated at the i-th location. In polar coordinates, it is given by

$$H_i^{(n,l)} = \cos\left(2\pi n r_i^2 + l\,\theta_i\right), \tag{3}$$

where r_i is the normalized radial distance and θ_i is the angular direction of the pixel. The integers $n \in \mathbb{Z}^+$ and $l \in \mathbb{Z}$ control the radial and angular frequency of the harmonic response, respectively.

The polar coordinates are discretized on a regular grid as

$$r_i^2 = \frac{u_i^2 + v_i^2}{k^2 + \epsilon}, \qquad \theta_i = \arctan 2\big(v_i,\, u_i + \epsilon\big), \tag{4}$$

where (u_i, v_i) are the offsets of the i-th pixel from the patch center, $k = \lfloor N/2 \rfloor$ and ϵ is a small constant for numerical stability.

To capture more complex and realistic symmetry patterns, we combine multiple harmonic orders. Let $\mathcal{O} = \{(n_1, l_1), \ldots, (n_M, l_M)\}$ be the selected set of orders, and let $\alpha_{n,l}$ be trainable coefficients that control their contribution. The final S-PDC response is then obtained by

$$y = \sum_{(n,l)\in\mathcal{O}} \alpha_{n,l} \sum_{i\neq c}^{N^2} w_i \cos(2\pi n r_i^2 + l\theta_i)(x_i - q). \tag{5}$$

By learning a sparse combination of harmonic orders through $\alpha_{n,l}$, S-PDC emphasizes symmetry-consistent structures and suppresses asymmetrical noise.

Ideally, for an input pattern x that exhibits K-fold rotational or reflectional symmetry, the proposed S-PDC produces almost identical responses under the corresponding symmetry transform. Let $\mathcal{S}_k$ denote a symmetry operator (e.g., a rotation of $2\pi/K$ or a reflection) belonging to the symmetry group $\mathcal{G}_{\text{sym}}$. Then S-PDC approximately satisfies

$$\text{S-PDC}(\mathcal{S}_k x) \approx \text{S-PDC}(x), \qquad \mathcal{S}_k \in \mathcal{G}_{\text{sym}}. \tag{6}$$

3.2. Rotation Invariant Symmetry Aware Pixel Difference Convolution (RIS-PDC)

Building upon S-PDC that models structural symmetry, RIS-PDC further integrates SO(2) rotation averaging to achieve full rotation invariance. The Lie Group provides a rigorous mathematical foundation for describing continuous and differentiable transformation groups. A special case of this is the $\mathrm{SO}(2)$ group, which represents all planar rotation matrices[1]. For any rotation angle θ, the rotation matrix can be expressed as formula (7):

$$R_\theta = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}, \tag{7}$$

which represents a counterclockwise rotation in 2D space by angle θ.

In RIS-PDC, the rotation operation is applied to the convolution kernels rather than to the feature maps themselves. For each sampled rotation angle θ, a rotated kernel $\boldsymbol{K}_\theta = R_\theta(\boldsymbol{K})$ is generated, while the input feature

[1]We implement discrete Group Convolution on the cyclic group C_n, achieving invariance by pooling over the kernel's rotation orbit. "Standard rotations" in the ablation refer to grid-preserving $k \cdot 90°$ steps.

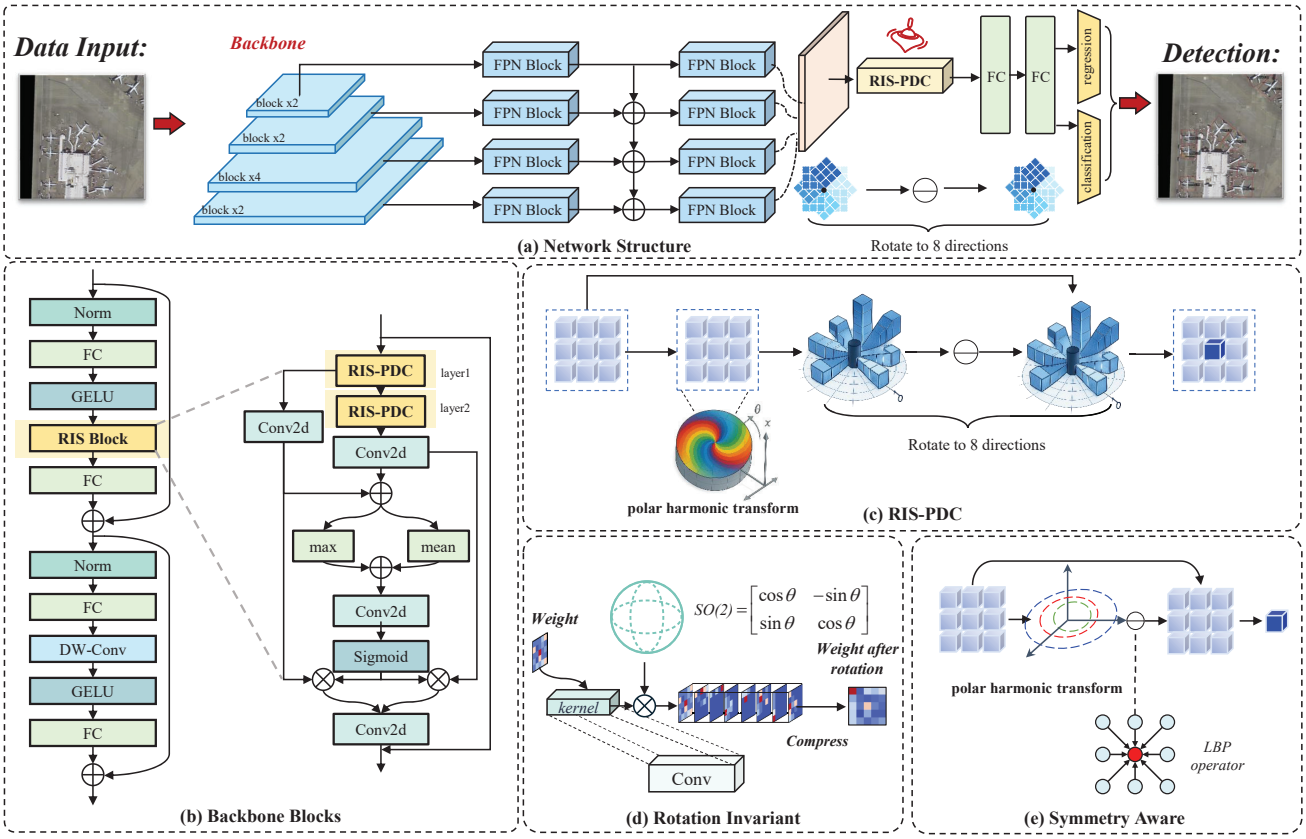


Figure 3. (a) The network structure of the proposed RIS-PiDiNet. (b) The architecture of the backbone block is outlined. In RIS-PiDiNet, the backbone consists chiefly of multiple backbone blocks. (c) The proposed RIS-PDC, which utilizes SO(2) for eight-direction rotations and employs the harmonic kernels to extract symmetry features. (d) The effect of eight-directional rotations on the convolution weights. (e) The application of the harmonic kernels and LBP operator to convolution.

$\boldsymbol{x}$ remains unchanged. The convolution responses under different rotated kernels are then aggregated to construct a rotation-invariant representation.

Formally, the response corresponding to angle θ can be written as:

$$y_\theta = (\boldsymbol{K}_\theta * \boldsymbol{x}) = (R_\theta \boldsymbol{K}) * \boldsymbol{x}, \tag{8}$$

where "*" denotes convolution. Assuming R_θ is linear, the equivariant property can be expressed as:

$$y_\theta = R_\theta(\boldsymbol{K} * \boldsymbol{x}) = R_\theta y. \tag{9}$$

To achieve rotation invariance, the responses across n discrete rotation angles $\{\theta_1, \theta_2, ..., \theta_n\}$ are averaged as:

$$y_{\text{final}} = \frac{1}{n}\sum_{j=1}^{n} y_{\theta_j} = \frac{1}{n}\sum_{j=1}^{n}(R_{\theta_j}\boldsymbol{K}) * \boldsymbol{x}. \tag{10}$$

We set n=8 (Config D), providing sufficient angular sampling to approximate SO(2) symmetry while minimizing interpolation artifacts on discrete grids. Fewer directions under-sample the orientation space; finer discretizations (e.g., n=16) yield diminishing returns (see Sec. 4.4).

This kernel-domain averaging suppresses orientation-dependent variations and yields a stable representation that remains invariant to input rotations, as shown in Eq. 11:

$$RIS\text{-}PDC(R_\phi \boldsymbol{x}) \approx RIS\text{-}PDC(\boldsymbol{x}). \tag{11}$$

where R_ϕ denotes a rotation of the input by angle ϕ.

4. Experiment

4.1. Datasets

We conduct extensive experiments on three popular remote sensing object detection datasets. DOTA-v1.0 [52] dataset is a large-scale dataset containing 2,806 images, 188,282 instances, and 15 categories. FAIR1M-v1.0 [45] is a remote sensing dataset consisting of 15,266 images, 1 million instances, and 37 categories. HRSC2016 [27] dataset contains 1,061 aerial images and 2,976 instances, while DIOR-R [20] dataset consists of 23,463 images and 192,518 instances. These datasets allow us to thoroughly evaluate the effectiveness and generalizability of our proposed method.

Table 1. Compared with the most advanced methods on the DOTA-v1.0 dataset, we trained our model on the trainval dataset and conducted tests on the test dataset, utilizing multi-scale training techniques. Pre.: "IN" represents pretrained weights derived from ImageNet1K, "CO" represents those from COCO2017, and "MA" represents those from MillionAID. RI.: "✓" indicates that the proposed method is designed to address rotation invariant/rotation sensitive characteristics, whereas "✗" denotes its absence. Category abbreviations: PL=Plane, BD=Baseball Diamond, BR=Bridge, GTF=Ground Track Field, SV=Small Vehicle, LV=Large Vehicle, SH=Ship, TC=Tennis Court, BC=Basketball Court, ST=Storage Tank, SBF=Soccer-ball Field, RA=Roundabout, HA=Harbor, SP=Swimming Pool, HC=Helicopter.

Method	Pre.	RI.	mAP ↑	#P	FLOPs	PL	BD	BR	GTF	SV	LV	SH	TC	BC	ST	SBF	RA	HA	SP	HC
Single-Scale																				
R3Det [59]	IN	✓	71.69	41.9M	336G	89.54	81.99	48.46	62.52	70.48	74.29	77.54	90.80	81.39	83.54	61.97	59.82	65.44	67.46	60.05
CFA [11]	IN	✗	75.05	-	-	89.26	81.72	51.81	67.17	79.99	78.25	84.46	90.77	82.40	85.54	54.86	67.75	73.04	70.24	64.96
SASM [14]	IN	✗	74.92	-	-	86.42	78.97	52.47	69.84	77.30	75.99	86.72	90.89	82.63	85.66	60.13	68.25	73.98	72.22	62.37
AO2-DETR [6]	IN	✓	77.73	74.3M	304G	89.27	84.97	56.67	74.89	78.87	82.73	87.35	90.50	84.68	85.41	61.97	69.96	74.68	72.39	**71.62**
S^2ANet [12]	IN	✓	76.11	-	-	88.70	81.41	54.28	69.75	78.04	80.54	88.04	90.69	84.75	86.22	65.03	65.81	76.16	73.37	58.86
R3Det-KLD [61]	IN	✗	77.36	41.9M	336G	89.90	84.17	55.80	69.35	78.72	84.08	87.00	89.75	84.32	85.73	64.74	61.80	76.62	**78.49**	70.89
SCRDet [58]	IN	✗	72.61	-	-	89.98	80.65	52.09	68.36	68.36	60.32	72.41	90.85	87.94	**86.86**	65.02	66.68	66.25	68.24	65.21
RoI Trans [7]	IN	✗	69.56	55.1M	200G	88.64	78.52	43.44	75.92	68.81	73.68	83.59	90.74	77.27	81.46	58.39	53.54	62.83	58.93	47.67
G.V. [55]	IN	✗	75.02	41.1M	198G	89.64	85.00	52.26	77.34	73.01	73.14	86.82	90.74	79.02	86.81	59.55	**70.91**	72.94	70.86	57.32
CenterMap [48]	IN	✗	71.59	41.1M	198G	89.02	80.56	49.41	61.98	77.99	74.19	83.74	89.44	78.01	83.52	47.64	65.93	63.68	67.07	61.59
CSL [56]	IN	✓	76.17	37.4M	236G	**90.25**	**85.53**	54.64	75.31	70.44	73.51	77.62	90.84	86.15	86.69	**69.60**	68.04	73.83	71.10	68.93
ReDet [13]	IN	✓	76.25	-	-	88.79	82.64	53.97	74.00	78.13	84.06	88.04	90.89	**87.78**	85.75	61.76	60.39	75.96	68.07	63.59
DODet [4]	IN	✗	75.89	-	-	89.61	83.10	51.43	72.02	79.16	81.99	87.71	90.89	86.53	84.56	62.21	65.38	71.98	70.79	61.93
AOPG [3]	IN	✗	75.39	-	-	89.14	82.74	51.87	69.28	77.65	82.42	88.08	90.89	86.26	85.13	60.60	66.30	74.05	67.76	58.77
O-RCNN [53]	IN	✗	74.68	41.1M	199G	89.71	79.65	49.80	70.59	77.85	76.95	87.45	90.89	85.83	83.95	59.28	65.56	66.35	70.63	65.82
LSKNet-T [23]	IN	✗	74.83	21.0M	124G	89.88	75.08	51.01	72.44	78.23	82.82	88.04	**90.90**	85.97	84.13	60.05	63.62	67.51	69.11	63.64
LSKNet-S [23]	IN	✗	77.49	31.0M	161G	89.66	85.52	**57.72**	75.70	74.95	78.69	88.24	90.88	86.79	86.38	66.92	63.77	**77.77**	74.47	64.82
MA3E [25]	MA	✓	77.93	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
PSC [64]	IN	✓	78.10	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
PKINet-S [1]	IN	✗	78.39	30.8 M	190 G	89.72	84.20	55.81	77.63	**80.25**	84.45	88.12	90.88	87.57	86.07	66.86	70.23	77.47	73.62	62.94
RIS-PiDiNet-T(**ours**)	IN	✓	76.92	21.0M	159G	89.48	81.99	52.46	75.65	78.66	83.51	88.06	90.88	87.03	85.37	65.11	65.80	74.05	71.47	64.33
RIS-PiDiNet-S(**ours**)	IN	✓	**78.53**	31.0M	206G	89.68	83.01	55.41	**77.69**	79.92	**85.10**	**88.27**	90.87	87.74	85.52	68.46	67.97	76.70	78.26	63.41
Multi-Scale																				
CSL [57]	IN	✓	76.17	37.4M	236G	**90.25**	85.53	54.64	75.31	70.44	73.51	77.62	90.84	86.15	86.69	69.60	68.04	73.83	71.10	68.93
CFA [11]	IN	✗	76.67	-	-	89.08	83.20	54.37	66.87	81.23	80.96	87.17	90.21	84.32	86.09	52.34	69.94	75.52	80.76	67.96
DAFNet [69]	IN	✗	76.95	-	-	89.40	86.27	53.70	60.51	82.04	81.17	88.66	90.37	83.81	87.27	53.93	69.38	75.61	81.26	70.86
DODet [2]	IN	✗	80.62	-	-	89.96	85.52	58.01	81.22	78.71	85.46	88.59	90.89	87.12	87.80	70.50	71.54	82.06	77.43	74.47
AOPG [3]	IN	✗	80.66	-	-	89.88	85.57	60.90	81.51	78.70	85.29	88.85	90.89	87.60	87.65	71.66	68.69	82.31	77.32	73.10
KFIoU [62]	IN	✗	80.93	58.8M	206G	89.44	84.41	62.22	82.51	80.10	86.07	88.68	90.90	87.32	**88.38**	72.80	71.95	78.96	74.95	75.27
RTMDet-R [28]	CO	✗	81.33	52.3M	205G	88.01	86.17	58.54	82.44	81.30	84.82	88.71	90.89	**88.77**	87.37	71.96	71.18	81.23	81.40	77.13
RVSA [47]	MA	✓	81.24	114.4M	414G	88.97	85.76	61.46	81.27	79.98	85.31	88.30	90.84	85.06	87.50	66.77	**73.11**	**84.75**	81.88	77.58
O-RCNN [53]	IN	✗	80.87	41.1M	199G	89.84	85.43	61.09	79.82	79.71	85.35	88.82	90.88	86.68	87.73	72.21	70.80	82.42	78.18	74.11
LSKNet-T [23]	IN	✗	81.37	21.0M	124G	89.14	84.90	61.78	83.50	81.54	85.87	88.64	90.89	88.02	87.31	71.55	70.74	78.66	79.81	**78.16**
LSKNet-S [23]	IN	✗	81.64	31.0M	161G	89.57	**86.34**	**63.13**	**83.67**	**82.20**	86.10	88.66	90.89	88.41	87.42	71.72	69.58	78.88	81.77	76.52
RIS-PiDiNet-T(**ours**)	IN	✓	81.52	21.0M	159G	89.44	84.96	62.41	83.23	81.32	85.94	88.79	90.95	87.17	87.93	74.25	70.06	78.96	82.25	75.14
RIS-PiDiNet-S(**ours**)	IN	✓	**81.81**	31.0M	206G	89.76	85.26	62.63	83.53	81.61	**86.25**	**89.11**	**91.28**	87.48	88.25	**74.52**	70.31	79.24	**82.55**	75.41

4.2. Implementation Details

The training pipeline consists of ImageNet-1K pretraining followed by fine-tuning on remote sensing detection datasets. In ablation studies, all models were trained for 100 epochs with a learning rate of 0.0005, batch size 512, and drop-path rate 0.1, while the final models were trained for 300 epochs. Following prior work, the training epochs for DOTA-v1.0, HRSC2016, and DIOR-R were set to 12, 36, and 12, with learning rates of 0.0002, 0.0004, and 0.0001, respectively. DOTA images were cropped into 1024×1024 patches with 200–500 pixel overlap, consistent with LSKNet preprocessing. AdamW was used with $\beta_1 = 0.9$, $\beta_2 = 0.999$, and weight decay 0.05. Pretraining was conducted on $8\times$ RTX 4090D GPUs; fine-tuning and testing were performed on A800-80GB GPUs (2 GPUs for training, batch size 8; 1 GPU for inference).

4.3. Main Results

Results on DOTA-v1.0. As shown in Table 1, RIS-PiDiNet-S delivers 1.04% improvement over LSKNet-S in single scale with minimal extra computation.

Table 2. Comparison with state-of-the-art methods on the HRSC2016 dataset. mAP(07) [9] and mAP(12) [8] respectively represent the metrics for VOC 2007 and VOC 2012.

Method	mAP(07)	mAP(12) ↑	#P	FLOPs
DRN [36]	-	92.70	-	-
CenterMap [48]	-	92.80	41.1M	198G
RoI Trans. [7]	86.20	-	55.1M	200G
G. V. [55]	88.20	-	41.1M	198G
R3Det [59]	89.26	96.01	41.9M	198G
DAL [32]	89.77	-	36.4M	216G
GWD [60]	89.85	97.37	47.4M	456G
S2ANet [12]	90.17	95.01	38.6M	198G
AOPG [3]	90.34	96.22	-	-
ReDet [13]	90.46	97.63	31.6M	-
O-RCNN [53]	90.50	97.60	41.1M	199G
LSKNet-T [23]	90.54	98.13	21.0M	124G
LSKNet-S [23]	90.65	98.46	31.0M	161G
PKINet-S [1]	90.65	98.54	30.8M	190G
RIS-PiDiNet-S	90.65	**98.60**	31.0M	206G

Results on HRSC2016. As shown in Table 2, RIS-PiDiNet-S achieves 98.60% mAP, surpassing all compared methods.

Results on DIOR-R. As shown in Table 3, RIS-PiDiNet-S achieves the best mAP of 67.28%.

Table 3. Comparison with state-of-the-art methods on the DIOR-R dataset.

Method	Pre.	#P	FLOPs	mAP ↑
RetinaNet-O [26]	IN	-	-	57.55
Faster RCNN-O [41]	IN	41.1M	198G	59.54
TIOE-Det [33]	IN	41.1M	198G	61.98
O-RCNN [53]	IN	41.1M	199G	64.30
ARS-DETR [66]	IN	41.1M	198G	66.12
O-RepPoints [21]	IN	36.6M	-	66.71
DCFL [54]	IN	-	-	66.80
LSKNet-S [23]	IN	31.0M	161G	65.90
PKINet-S [1]	IN	30.8M	190G	67.03
MA3E [25]	IN	-	-	66.73
RIS-PiDiNet-S	IN	31.0M	206G	**67.28**

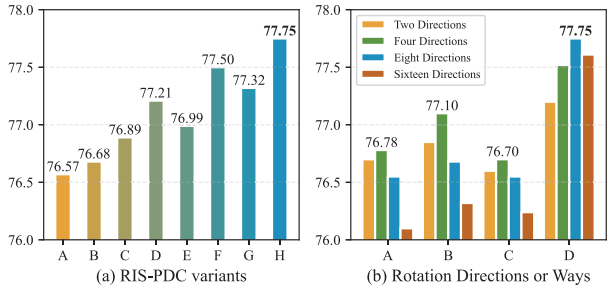


Figure 4. Ablation studies were performed on both RIS-PDC variants and the effect of rotation directions. (a) the performance of different RIS-PDC variants is evaluated. (b) is the influence of rotation direction.

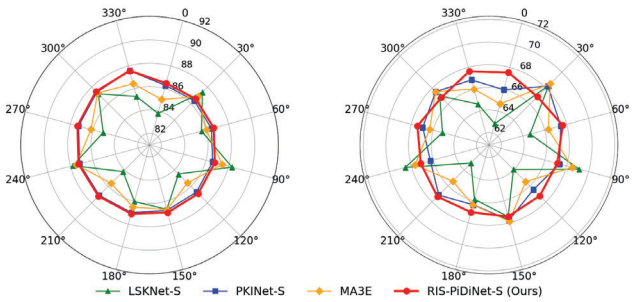


Figure 5. (a) Detection performance of four different models on DOTAv1.0 val dataset across multiple rotation angles. (b) Detection performance of three different models on DIOR test dataset across multiple rotation angles. These plots illustrate the models' robustness to variations in object orientation.

4.4. Ablation

In this section, we present the results of ablation studies conducted on the single scale of the DOTA-v1.0 test set.

Effectiveness of applications at different layers. As shown in Table 4, inserting RIS-PDC at layer1, layer2, and the detection head improves performance with minimal cost and no extra parameters. Notably, removing layer2 yields 77.60% mAP while saving 42G FLOPs (layer1 + head), offering a favorable accuracy-efficiency trade-off.

Ablation of RIS-PDC variants. PDC enhances small aerial targets as a high-frequency filter; S-PDC targets structural symmetry under occlusion; RIS-PDC integrates



Figure 6. Detection outputs of LSKNet-S, PKINet-S, MA3E, and RIS-PiDiNet-S on multiple remote sensing images. Each row corresponds to one input sample, and each column corresponds to the prediction of a different model.

Table 4. The influence of using RIS-PDC in different network layers. "✓" indicates that RIS-PDC is employed at that position, while "✗" denotes its absence.

layer1	layer2	layer_head	#P	FLOPs	mAP ↑
✗	✗	✗	31.0M	161G	76.57
✓	✗	✗	31.0M	164G	76.79
✗	✓	✗	31.0M	203G	76.65
✗	✗	✓	31.0M	161G	77.41
✓	✓	✗	31.0M	206G	76.84
✓	✗	✓	31.0M	164G	77.60
✗	✓	✓	31.0M	203G	77.52
✓	✓	✓	31.0M	206G	77.75

SO(2) priors for global rotation invariance. We explore multiple convolutional variants for the RPN head's first layer: "A" Vanilla Conv, "B" PDC only, "C" PHT+Conv, "D" SO(2)+Conv, "E" S-PDC, "F" SO(2)+PDC, "G" PHT+SO(2)+Conv, "H" RIS-PDC. As shown in Figure 4(a), RIS-PDC achieves the best mAP.

Effectiveness of different rotation directions. We ablate rotation strategies and directional sampling (Figure 4(b)), comparing standard 2D discrete rotations, rotation without group structure, kernel sorting, and SO(2) under 2, 4, 8, and 16 orientations. SO(2) with 8 directions achieves the best performance.

Symmetry aware module setting. Table 5 summarizes the ablation of S-PDC. A1 and A2 denote standard convolution and PDC with LBP, yielding 76.57% and 76.68% mAP. Adding a single harmonic order (A3, A4) brings small gains, with polar mapping giving a slight boost. Expanding

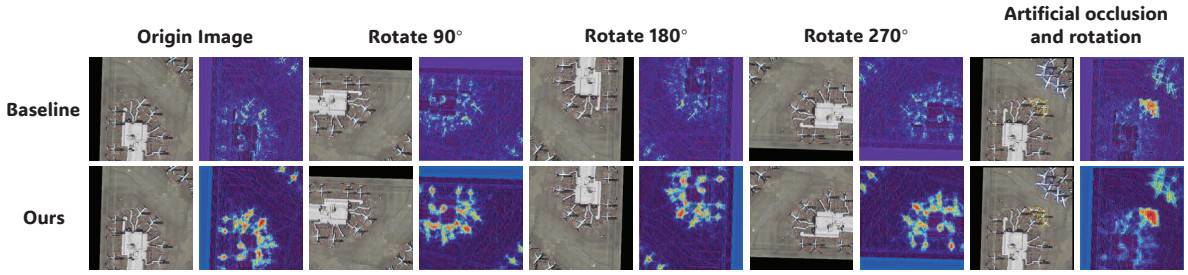


Figure 7. Detection results and corresponding feature maps for the Baseline (LSKNet-S) and our method (RIS-PiDiNet-S) are presented on two inputs: the original image and a artificial modified version featuring one-sided occlusion of the symmetric structure coupled with object rotation.

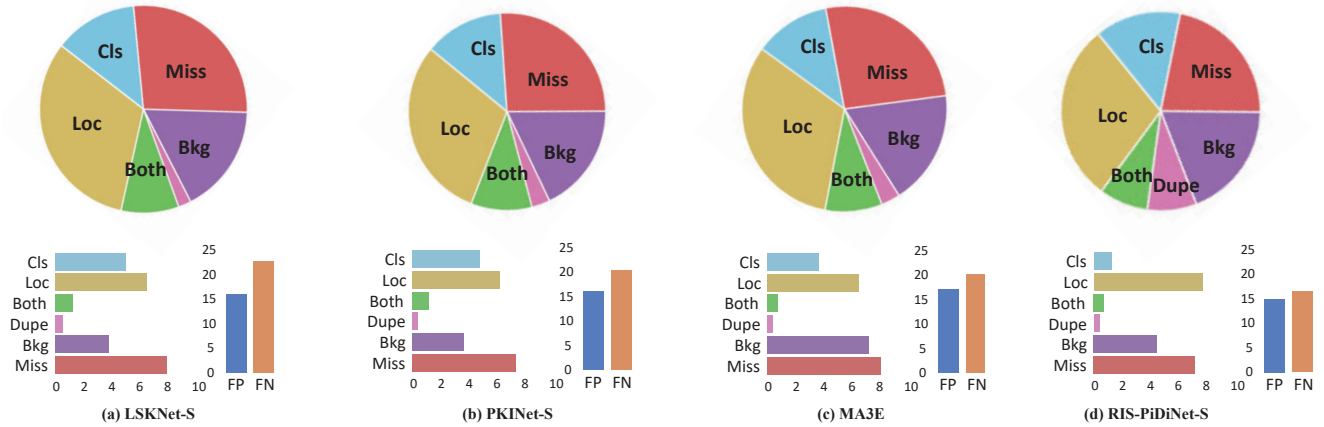


Figure 8. TIDE error distribution comparison across four detectors: LSKNet-S, PKINet-S, MA3E, and the proposed RIS-PiDiNet-S. Each pie chart shows the proportion of six error types (Cls, Loc, Both, Dupe, Bkg, Miss), and the bar plots summarize the corresponding FP and FN contributions under the TIDE definition.

Table 5. Ablation study on harmonic kernel configuration and polar coordinate modeling.

Setting	Harmonic (n, l)	#P	FLOPs	mAP ↑
A1	–	31.0M	198G	76.57
A2	–	31.0M	198G	76.68
A3	$(1, 0)$	31.0M	202G	76.92
A4	$(1, 0)$	31.0M	202G	76.73
A5	$\{(1, 0), (2, 1), (3, 2)\}$	31.0M	204G	77.23
A6	$\{(1, 0), (2, 1), (3, 2)\}$	31.0M	204G	76.95
A7	Learned	31.0M	206G	**77.75**
A8	Learned	31.0M	210G	77.31
A9	Learned	31.6M	216G	77.48

to three fixed harmonic orders (A5, A6) further improves results, again with better performance under polar coordinates. The learnable harmonic setting in A7 achieves the best accuracy (77.75%), while removing polar terms (A8) leads to a small drop. Increasing to five learnable orders (A9) offers only marginal benefit, indicating a saturation point. Overall, the results show that harmonic modulation and polar encoding work jointly to capture structural symmetry effectively.

Ablation on symmetry aware. To evaluate different symmetry aware strategies in RIS-PDC, we compare the proposed S-PDC with Laplacian operator, Hu/Zernike moments, Gabor filters, and steerable filters. As shown in Table 6, RIS-PDC achieves the highest mAP of 78.53% with minimal computational and parameter overhead, highlighting the superiority of the proposed approach.

Table 6. The influence of different symmetry modeling methods integrated into RIS-PDC. All variants share the same network architecture.

Method	#P	FLOPs	mAP ↑
Laplacian	31.0M	206G	76.98
Hu/Zernike Moments	31.0M	206G	75.12
Gabor Filters	31.0M	216G	76.10
Steerable Filters	32.1M	211G	77.05
RIS-PDC (Ours)	31.0M	206G	**77.75**

Performance of RIS-PiDiNet backbone under different detection frameworks. To validate the generality and efficiency of the proposed RIS-PiDiNet backbone, we evaluate it alongside ResNet-18 and LSKNet-T under multiple remote sensing detection frameworks, including both one-stage (YOLO) and two-stage (O-RCNN, RoI Transformer, S^2A-Net, R3Det) paradigms. As summarized in Table 7, replacing ResNet-18 with LSKNet-T leads to clear accu-

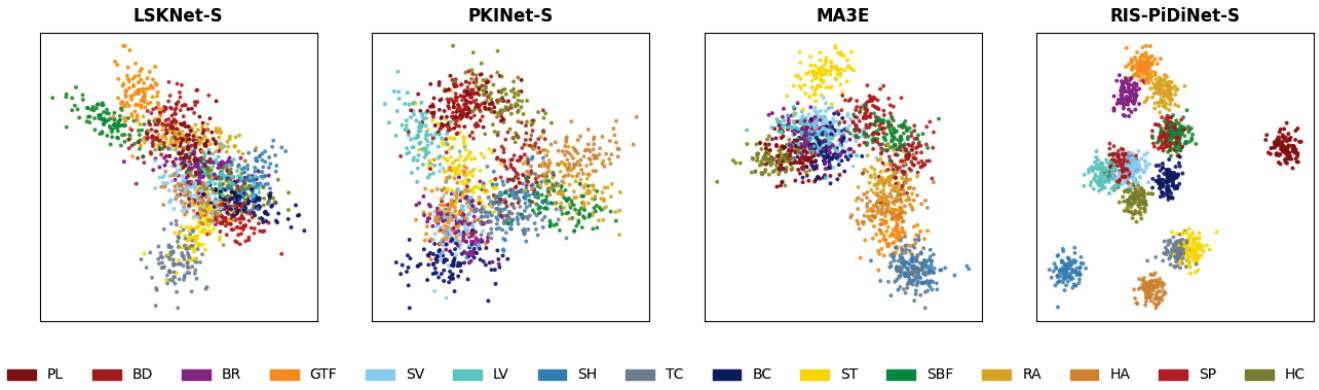


Figure 9. t-SNE visualization of feature embeddings from four detectors (LSKNet-S, PKINet-S, MA3E, and RIS-PiDiNet-S) on the DOTA dataset. Each point corresponds to a detected instance, colored by object category.

racy gains across all frameworks while cutting the parameter count from 11.2M to 4.3M and halving the FLOPs from 38.1G to 19.1G. Building on this efficiency, our RIS-PiDiNet-T further improves the overall mAP by approximately 2 points on average and achieves the best performance under all frameworks, while maintaining the same lightweight model size (4.3M parameters) and moderate computation cost of 54.1 GFLOPs. These results demonstrate that RIS-PiDiNet-T achieves an excellent balance between accuracy and efficiency.

Table 7. Comparison of detection performance under different frameworks using ResNet-18, LSKNet-T, and RIS-PiDiNet-T backbones. RIS-PiDiNet-T consistently achieves the highest accuracy while remaining lightweight.

Frameworks	ResNet-18	LSKNet-T	RIS-PiDiNet-T
YOLO [40]	71.85	73.92	75.86
O-RCNN [53]	72.31	74.83	76.92
RoI Trans. [7]	71.68	74.41	76.33
S^2A-Net [12]	70.27	73.85	75.79
R3Det [59]	69.42	72.94	74.96
#P (backbone only)	11.2M	4.3M	4.3M
FLOPs (backbone only)	38.1G	19.1G	54.1G

4.5. Exploration of Angle Robustness

As shown in Figure 5, RIS-PiDiNet-S maintains nearly uniform accuracy across all rotation angles, whereas LSKNet-S, PKINet-S, and MA3E show larger directional fluctuations, confirming that our design effectively preserves geometric consistency under arbitrary orientations.

4.6. Visualization

Detection results. In Figures 6 and 7, we present both detection and feature-level comparisons to comprehensively evaluate the rotation-invariant capability of the proposed RIS-PiDiNet-S. Figure 6 visualizes detection results and intermediate feature maps on both the original and artificially occluded inputs. Compared with LSKNet-S, PKINet-S, and MA3E, our model produces clearer activation responses along symmetric boundaries and maintains geometric consistency even under one-sided occlusion, demonstrating its robustness against structural degradation. Figure 7 further analyzes rotation consistency by rotating the same image by 90°, 180°, and 270°, as well as introducing artificial occlusion combined with rotation. The baseline model (LSKNet-S) exhibits strong orientation sensitivity and unstable activation shifts, whereas RIS-PiDiNet-S maintains stable, rotation-equivariant responses across all transformations.

TIDE error analysis. Figure 8 shows that RIS-PiDiNet-S achieves the lowest Miss and Loc errors with a balanced distribution, confirming improved geometric stability over LSKNet-S, PKINet-S, and MA3E.

t-SNE analysis. Figure 9 shows that RIS-PiDiNet-S forms more compact and clearly separated clusters than LSKNet-S, PKINet-S, and MA3E, indicating more stable features and better inter-class separability.

5. Conclusion

In this study, we leveraged Lie group rotations to adapt convolution weights and apply weighted features, thereby achieving rotation invariance without increasing parameters. To handle symmetry and local feature enhancement, we employed the polar harmonic transform and LBP-based differentials, which together facilitate robust symmetric feature extraction. We integrate these innovations into a unified framework and propose RIS-PiDiNet. Extensive experiments show that this method exhibits strong generalization capability and achieves state-of-the-art performance on DOTA-v1.0, HRSC2016, and DIOR-R benchmarks. We hope this research will provide new insights into the development of the remote sensing field by advancing rotation-invariant and symmetry-aware feature modeling.

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References

[1] Xinhao Cai, Qiuxia Lai, Yuwei Wang, Wenguan Wang, Zeren Sun, and Yazhou Yao. Poly kernel inception network for remote sensing detection. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 27706–27716, 2024. 2, 5, 6

[2] Jiale Cao, Hisham Cholakkal, Rao Muhammad Anwer, Fahad Shahbaz Khan, Yanwei Pang, and Ling Shao. D2det: Towards high quality object detection and instance segmentation. In *IEEE Conf. Comput. Vis. Pattern Recog.*, pages 11485–11494, 2020. 5

[3] Gong Cheng, Jiabao Wang, Ke Li, Xingxing Xie, Chunbo Lang, Yanqing Yao, and Junwei Han. Anchor-free oriented proposal generator for object detection. *IEEE Trans. Geosci. Remote Sens.*, 60:1–11, 2022. 5

[4] Gong Cheng, Yanqing Yao, Shengyang Li, Ke Li, Xingxing Xie, Jiabao Wang, Xiwen Yao, and Junwei Han. Dual-aligned oriented detector. *IEEE Trans. Geosci. Remote Sens.*, 60:1–11, 2022. 5

[5] Taco Cohen and Max Welling. Group equivariant convolutional networks. In *International conference on machine learning*, pages 2990–2999. PMLR, 2016. 2

[6] Linhui Dai, Hong Liu, Hao Tang, Zhiwei Wu, and Pinhao Song. Ao2-detr: Arbitrary-oriented object detection transformer. *IEEE Trans. Circuits Syst. Video Technol.*, 2022. 2, 5

[7] Jian Ding, Nan Xue, Yang Long, Gui-Song Xia, and Qikai Lu. Learning roi transformer for oriented object detection in aerial images. In *IEEE Conf. Comput. Vis. Pattern Recog.*, pages 2849–2858, 2019. 2, 5, 8

[8] Mark Everingham and John Winn. The pascal visual object classes challenge 2012 (voc2012) development kit. *Pattern Analysis, Statistical Modelling and Computational Learning, Tech. Rep*, 8(5):2–5, 2011. 5

[9] Mark Everingham, Luc Van Gool, Christopher KI Williams, John Winn, and Andrew Zisserman. The pascal visual object classes (voc) challenge. *International journal of computer vision*, 88:303–338, 2010. 5

[10] Hans Freudenthal. Lie groups in the foundations of geometry. *Advances in Mathematics*, 1(2):145–190, 1964. 2, 3

[11] Zonghao Guo, Chang Liu, Xiaosong Zhang, Jianbin Jiao, Xiangyang Ji, and Qixiang Ye. Beyond bounding-box: Convex-hull feature adaptation for oriented and densely packed object detection. In *IEEE Conf. Comput. Vis. Pattern Recog.*, pages 8792–8801, 2021. 5

[12] Jiaming Han, Jian Ding, Jie Li, and Gui-Song Xia. Align deep features for oriented object detection. *IEEE Trans. Geosci. Remote Sens.*, 60:1–11, 2021. 2, 5, 8

[13] Jiaming Han, Jian Ding, Nan Xue, and Gui-Song Xia. Redet: A rotation-equivariant detector for aerial object detection. In *IEEE Conf. Comput. Vis. Pattern Recog.*, pages 2786–2795, 2021. 2, 5

[14] Liping Hou, Ke Lu, Jian Xue, and Yuqiu Li. Shape-adaptive selection and measurement for oriented object detection. In *AAAI*, pages 923–932, 2022. 5

[15] Zibo Hu, Kun Gao, Xiaodian Zhang, Junwei Wang, Hong Wang, Zhijia Yang, Chenrui Li, and Wei Li. Emo2-detr: Efficient-matching oriented object detection with transformers. *IEEE Trans. Geosci. Remote Sens.*, 2023. 2

[16] Tao Huang, Xiaohuan Pei, Shan You, Fei Wang, Chen Qian, and Chang Xu. Localmamba: Visual state space model with windowed selective scan. In *European conference on computer vision*, pages 12–22. Springer, 2024. 3

[17] Ernie G Kalnins and Willard Miller Jr. Lie theory and separation of variables. 4. the groups so (2, 1) and so (3). *Journal of Mathematical Physics*, 15(8):1263–1274, 1974. 2

[18] Chanho Lee, Jinsu Son, Hyounguk Shon, Yunho Jeon, and Junmo Kim. Fred: Towards a full rotation-equivariance in aerial image object detection. In *Proceedings of the AAAI Conference on Artificial Intelligence*, pages 2883–2891, 2024. 2

[19] Haibin Li, En Tian, Wenming Zhang, Yaqian Li, and Junteng Cao. Improving remote sensing object detection by using feature extraction and rotational equivariant attention. *International Journal of Remote Sensing*, 45(11):3789–3806, 2024. 2

[20] Ke Li, Gang Wan, Gong Cheng, Liqiu Meng, and Junwei Han. Object detection in optical remote sensing images: A survey and a new benchmark. *ISPRS J. Photogramm. Remote Sens.*, 159:296–307, 2020. 1, 4

[21] Wentong Li, Yijie Chen, Kaixuan Hu, and Jianke Zhu. Oriented reppoints for aerial object detection. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 1829–1838, 2022. 6

[22] Xiang Li, Zixuan Huang, Anh Thai, and James M Rehg. Symmetry strikes back: From single-image symmetry detection to 3d generation. *arXiv preprint arXiv:2411.17763*, 2024. 2

[23] Yuxuan Li, Qibin Hou, Zhaohui Zheng, Ming-Ming Cheng, Jian Yang, and Xiang Li. Large selective kernel network for remote sensing object detection. In *Proceedings of the IEEE/CVF international conference on computer vision*, pages 16794–16805, 2023. 2, 3, 5, 6

[24] Zheng Li, Yongcheng Wang, Ning Zhang, Yuxi Zhang, Zhikang Zhao, Dongdong Xu, Guangli Ben, and Yunxiao Gao. Deep learning-based object detection techniques for remote sensing images: A survey. *Remote Sensing*, 14(10): 2385, 2022. 2

[25] Zhihao Li, Biao Hou, Siteng Ma, Zitong Wu, Xianpeng Guo, Bo Ren, and Licheng Jiao. Masked angle-aware autoencoder for remote sensing images. In *European Conference on Computer Vision*, pages 260–278. Springer, 2024. 5, 6

[26] Tsung-Yi Lin, Priya Goyal, Ross Girshick, Kaiming He, and Piotr Dollár. Focal loss for dense object detection. In *Proceedings of the IEEE international conference on computer vision*, pages 2980–2988, 2017. 2, 6

[27] Zikun Liu, Liu Yuan, Lubin Weng, and Yiping Yang. A high resolution optical satellite image dataset for ship recognition and some new baselines. In *International conference on pattern recognition applications and methods*, pages 324–331. SciTePress, 2017. 4

[28] Chengqi Lyu, Wenwei Zhang, Haian Huang, Yue Zhou, Yudong Wang, Yanyi Liu, Shilong Zhang, and Kai Chen. Rtmdet: An empirical study of designing real-time object detectors. *arXiv preprint arXiv:2212.07784*, 2022. 5

[29] Xinyu Ma, Pengyuan Lv, and Yanfei Zhong. Qetr: A query-enhanced transformer for remote sensing image object detection. *IEEE Geosci. Remote Sens. Lett.*, 21:1–5, 2024. 2

[30] Benjamin Midtvedt, Jesús Pineda, Fredrik Skärberg, Erik Olsén, Harshith Bachimanchi, Emelie Wesén, Elin K Esbjörner, Erik Selander, Fredrik Höök, Daniel Midtvedt, et al. Single-shot self-supervised object detection in microscopy. *Nature communications*, 13(1):7492, 2022. 2

[31] Qi Ming, Lingjuan Miao, Zhiqiang Zhou, Xue Yang, and Yunpeng Dong. Optimization for arbitrary-oriented object detection via representation invariance loss. *IEEE Geoscience and Remote Sensing Letters*, 19:1–5, 2021. 2

[32] Qi Ming, Zhiqiang Zhou, Lingjuan Miao, Hongwei Zhang, and Linhao Li. Dynamic anchor learning for arbitrary-oriented object detection. In *Proceedings of the AAAI conference on artificial intelligence*, pages 2355–2363, 2021. 5

[33] Qi Ming, Lingjuan Miao, Zhiqiang Zhou, Junjie Song, Yunpeng Dong, and Xue Yang. Task interleaving and orientation estimation for high-precision oriented object detection in aerial images. *ISPRS Journal of Photogrammetry and Remote Sensing*, 196:241–255, 2023. 6

[34] Indranil Misra, Mukesh Kumar Rohil, S Manthira Moorthi, and Debajyoti Dhar. Feature based remote sensing image registration techniques: A comprehensive and comparative review. *International Journal of Remote Sensing*, 43(12): 4477–4516, 2022. 2

[35] Alhassan Mumuni and Fuseini Mumuni. Cnn architectures for geometric transformation-invariant feature representation in computer vision: a review. *SN Computer Science*, 2(5): 340, 2021. 2

[36] Xingjia Pan, Yuqiang Ren, Kekai Sheng, Weiming Dong, Haolei Yuan, Xiaowei Guo, Chongyang Ma, and Changsheng Xu. Dynamic refinement network for oriented and densely packed object detection. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 11207–11216, 2020. 5

[37] Matti Pietikäinen. Local binary patterns. *Scholarpedia*, 5(3): 9775, 2010. 2, 3, 1

[38] Wen Qian, Xue Yang, Silong Peng, Junchi Yan, and Yue Guo. Learning modulated loss for rotated object detection. In *AAAI*, pages 2458–2466, 2021. 2

[39] Xiaoliang Qian, Baokun Wu, Gong Cheng, Xiwen Yao, Wei Wang, and Junwei Han. Building a bridge of bounding box regression between oriented and horizontal object detection in remote sensing images. *IEEE Transactions on Geoscience and Remote Sensing*, 61:1–9, 2023. 2

[40] Joseph Redmon and Ali Farhadi. Yolov3: An incremental improvement. *arXiv preprint arXiv:1804.02767*, 2018. 8

[41] Shaoqing Ren, Kaiming He, Ross Girshick, and Jian Sun. Faster r-cnn: Towards real-time object detection with region proposal networks. *Advances in neural information processing systems*, 28, 2015. 6

[42] Isaac Robinson, Peter Robicheaux, Matvei Popov, Deva Ramanan, and Neehar Peri. Rf-detr: neural architecture search for real-time detection transformers. *arXiv preprint arXiv:2511.09554*, 2025. 3

[43] Pourya Shamsolmoali, Masoumeh Zareapoor, Jocelyn Chanussot, Huiyu Zhou, and Jie Yang. Rotation equivariant feature image pyramid network for object detection in optical remote sensing imagery. *IEEE Transactions on Geoscience and Remote Sensing*, 60:1–14, 2021. 2

[44] Zhuo Su, Jiehua Zhang, Longguang Wang, Hua Zhang, Zhen Liu, Matti Pietikäinen, and Li Liu. Lightweight pixel difference networks for efficient visual representation learning. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 45(12):14956–14974, 2023. 2, 3

[45] Xian Sun, Peijin Wang, Zhiyuan Yan, Feng Xu, Ruiping Wang, Wenhui Diao, Jin Chen, Jihao Li, Yingchao Feng, Tao Xu, et al. Fair1m: A benchmark dataset for fine-grained object recognition in high-resolution remote sensing imagery. *ISPRS J. Photogramm. Remote Sens.*, 184:116–130, 2022. 4

[46] Yang Tian, Mengmeng Zhang, Jinyu Li, Yangfan Li, Hong Yang, and Wei Li. Fpnformer: Rethink the method of processing the rotation-invariance and rotation-equivariance on arbitrary-oriented object detection. *IEEE Transactions on Geoscience and Remote Sensing*, 62:1–10, 2024. 2

[47] Di Wang, Qiming Zhang, Yufei Xu, Jing Zhang, Bo Du, Dacheng Tao, and Liangpei Zhang. Advancing plain vision transformer toward remote sensing foundation model. *IEEE Trans. Geosci. Remote Sens.*, 61:1–15, 2022. 5

[48] Jinwang Wang, Wen Yang, Heng-Chao Li, Haijian Zhang, and Gui-Song Xia. Learning center probability map for detecting objects in aerial images. *IEEE Trans. Geosci. Remote Sens.*, 59(5):4307–4323, 2020. 2, 5

[49] Long Wen, Yu Cheng, Yi Fang, and Xinyu Li. A comprehensive survey of oriented object detection in remote sensing images. *Expert Systems with Applications*, 224:119960, 2023. 1

[50] Yue Wu, Yinpeng Chen, Lu Yuan, Zicheng Liu, Lijuan Wang, Hongzhi Li, and Yun Fu. Rethinking classification and localization for object detection. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 10186–10195, 2020. 2

[51] Zhiqiang Wu, Yingjie Liu, Hanlin Dong, Xuan Tang, Jian Yang, Bo Jin, Mingsong Chen, and Xian Wei. R2det: Exploring relaxed rotation equivariance in 2d object detection. In *The Thirteenth International Conference on Learning Representations*, 2025. 2

[52] Gui-Song Xia, Xiang Bai, Jian Ding, Zhen Zhu, Serge Belongie, Jiebo Luo, Mihai Datcu, Marcello Pelillo, and Liangpei Zhang. Dota: A large-scale dataset for object detection

in aerial images. In *Proceedings of the IEEE conference on computer vision and pattern recognition*, pages 3974–3983, 2018. 4

[53] Xingxing Xie, Gong Cheng, Jiabao Wang, Xiwen Yao, and Junwei Han. Oriented r-cnn for object detection. In *Int. Conf. Comput. Vis.*, pages 3520–3529, 2021. 2, 5, 6, 8

[54] Chang Xu, Jian Ding, Jinwang Wang, Wen Yang, Huai Yu, Lei Yu, and Gui-Song Xia. Dynamic coarse-to-fine learning for oriented tiny object detection. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 7318–7328, 2023. 6

[55] Yongchao Xu, Mingtao Fu, Qimeng Wang, Yukang Wang, Kai Chen, Gui-Song Xia, and Xiang Bai. Gliding vertex on the horizontal bounding box for multi-oriented object detection. *IEEE Trans. Pattern Anal. Mach. Intell.*, 43(4):1452–1459, 2020. 2, 5

[56] Xue Yang and Junchi Yan. Arbitrary-oriented object detection with circular smooth label. In *Eur. Conf. Comput. Vis.*, pages 677–694. Springer, 2020. 2, 5

[57] Xue Yang and Junchi Yan. On the arbitrary-oriented object detection: Classification based approaches revisited. *Int. J. Comput. Vis.*, 130(5):1340–1365, 2022. 2, 5

[58] Xue Yang, Jirui Yang, Junchi Yan, Yue Zhang, Tengfei Zhang, Zhi Guo, Xian Sun, and Kun Fu. Scrdet: Towards more robust detection for small, cluttered and rotated objects. In *Int. Conf. Comput. Vis.*, pages 8232–8241, 2019. 2, 5

[59] Xue Yang, Junchi Yan, Ziming Feng, and Tao He. R3det: Refined single-stage detector with feature refinement for rotating object. In *AAAI*, pages 3163–3171, 2021. 2, 5, 8

[60] Xue Yang, Junchi Yan, Qi Ming, Wentao Wang, Xiaopeng Zhang, and Qi Tian. Rethinking rotated object detection with gaussian wasserstein distance loss. In *International conference on machine learning*, pages 11830–11841. PMLR, 2021. 5

[61] Xue Yang, Xiaojiang Yang, Jirui Yang, Qi Ming, Wentao Wang, Qi Tian, and Junchi Yan. Learning high-precision bounding box for rotated object detection via kullback-leibler divergence. *Adv. Neural Inform. Process. Syst.*, 34: 18381–18394, 2021. 5

[62] Xue Yang, Yue Zhou, Gefan Zhang, Jirui Yang, Wentao Wang, Junchi Yan, Xiaopeng Zhang, and Qi Tian. The kfiou loss for rotated object detection. *arXiv preprint arXiv:2201.12558*, 2022. 5

[63] Pew-Thian Yap, Xudong Jiang, and Alex Chichung Kot. Two-dimensional polar harmonic transforms for invariant image representation. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 32(7):1259–1270, 2009. 2, 3, 1

[64] Yi Yu and Feipeng Da. Phase-shifting coder: Predicting accurate orientation in oriented object detection. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 13354–13363, 2023. 5

[65] Zitong Yu, Chenxu Zhao, Zezheng Wang, Yunxiao Qin, Zhuo Su, Xiaobai Li, Feng Zhou, and Guoying Zhao. Searching central difference convolutional networks for face anti-spoofing. In *IEEE Conf. Comput. Vis. Pattern Recog.*, pages 5295–5305, 2020. 3

[66] Ying Zeng, Yushi Chen, Xue Yang, Qingyun Li, and Junchi Yan. Ars-detr: Aspect ratio-sensitive detection transformer for aerial oriented object detection. *IEEE Trans. Geosci. Remote Sens.*, 62:1–15, 2024. 2, 6

[67] Jialei Zhan, Liang Bai, Jiehua Zhang, Tianpeng Liu, Fan Shi, Yongxiang Liu, and Li Liu. Heterogeneous binary pixel difference networks for remote sensing object detection. *IEEE Transactions on Geoscience and Remote Sensing*, 2024. 2, 3

[68] Gongjie Zhang, Shijian Lu, and Wei Zhang. Cad-net: A context-aware detection network for objects in remote sensing imagery. *IEEE Transactions on Geoscience and Remote Sensing*, 57(12):10015–10024, 2019. 2

[69] Qijian Zhang, Runmin Cong, Chongyi Li, Ming-Ming Cheng, Yuming Fang, Xiaochun Cao, Yao Zhao, and Sam Kwong. Dense attention fluid network for salient object detection in optical remote sensing images. *IEEE Trans. Image Process.*, 30:1305–1317, 2020. 5

[70] Xiangrong Zhang, Tianyang Zhang, Guanchun Wang, Peng Zhu, Xu Tang, Xiuping Jia, and Licheng Jiao. Remote sensing object detection meets deep learning: A metareview of challenges and advances. *IEEE Geoscience and Remote Sensing Magazine*, 11(4):8–44, 2023. 1, 2

[71] Yanzhao Zhou, Qixiang Ye, Qiang Qiu, and Jianbin Jiao. Oriented response networks. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, pages 519–528, 2017. 2

[72] Bai Zhu, Chao Yang, Jinkun Dai, Jianwei Fan, and Yuanxin Ye. R2fd2: Fast and robust matching of multimodal remote sensing image via repeatable feature detector and rotation-invariant feature descriptor. *arXiv preprint arXiv:2212.02277*, 2022. 2

[73] Kun Zhu, Xiaodong Zhang, Guanzhou Chen, Xianwei Li, Peihua Cai, Puyun Liao, and Tong Wang. Multi-oriented rotation-equivariant network for object detection on remote sensing images. *IEEE Geoscience and Remote Sensing Letters*, 19:1–5, 2022. 2